BLOCKCHAIN BASED ENERGY SHARING BETWEEN HOUSES

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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CERTIFICATE

This is to certify that the Use Case report entitled **"Blockchain Based Energy Sharing Between Houses"** that is being submitted by **KANAPARTHI AMALA (23501A0560)**, as part of Assignment-2 for the **Blockchain Technology**(**23CS4701B**) course in **4-1** during the academic year **2026-27**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across multiple nodes, ensuring transparency, security, and immutability. Unlike traditional databases, blockchain operates on a peer-to-peer network, eliminating the need for central authorities. Each block in the chain contains a cryptographic hash of the previous block, making data tampering virtually impossible. Originally designed for cryptocurrency transactions, blockchain has now expanded into various domains, including finance, supply chain, healthcare, and energy management. [1]

1.2. Relevance of Blockchain in Energy Sharing Between Houses

Modern households increasingly generate their own electricity through rooftop solar panels and other small renewable sources. A household that generates more electricity than it consumes has surplus energy, while a neighbouring household may at the same time need additional energy. In the traditional model, the surplus is fed back to the utility grid at a fixed tariff and the neighbour buys the same energy from the utility at a much higher price. The two households cannot trade directly because there is no simple, trusted way to record and settle such small transactions. [2]

A blockchain-based energy sharing system provides a tamper-resistant mechanism for recording energy offers, purchases and payments between houses. By leveraging smart contracts, offers can be created, matched and settled automatically without intermediaries, reducing processing time and human errors. Every energy transaction is stored as a permanent record that authorized households and administrators can verify at any time. [3]

The adoption of blockchain in local energy sharing enhances trust between neighbours, enabling a secure, transparent and fraud-resistant energy market. This transformation encourages households to install renewable generation, reduces energy wasted by unused surplus, and lowers dependence on a single central utility.

Furthermore, the decentralized nature of blockchain ensures that energy transaction data is immutable and not controlled by any single party, eliminating the risk of records being altered or lost at a central server. This feature is particularly beneficial where multiple parties, such as producers, consumers, grid operators and administrators, need access to the same trusted record. As more houses adopt rooftop solar and smart meters, blockchain-based energy sharing offers a scalable and future-ready approach to managing local energy markets efficiently and securely. [2] [3]

2. BACKGROUND

Traditional electricity distribution is built around one-way supply from large power plants to households through a centralized utility. As households become producers of energy, this model creates several inefficiencies and prevents surplus energy from being shared fairly. Below are some key issues with the existing systems:

2.1. Wastage of Surplus Energy

Households with solar panels often generate more electricity than they use during the day. Without a way to share it locally, the surplus is either exported to the grid at a low rate or wasted when storage is unavailable.

2.2. Unfair Pricing for Producers and Consumers

Utilities usually buy surplus energy from households at a low feed-in tariff and sell electricity to consumers at a higher retail tariff. The price difference benefits the intermediary rather than the neighbours who actually produce and use the energy. [2]

2.3. Dependence on Centralized Intermediaries

Energy trading normally requires a utility or broker to measure, bill and settle every transaction. This adds cost, delays and a single point of control over who can sell and buy energy.

2.4. Lack of Transparency and Traceability

Households have limited visibility into how much energy was exported, who consumed it and how the price was calculated. This lack of transparency creates billing disputes and reduces trust in the system. [1]

2.5. Risk of Data Tampering and Fraud

Centralized billing databases are vulnerable to cyberattacks, unauthorized modifications and internal manipulation. Incorrect or altered energy records can lead to financial loss for households and utilities.

2.6. Difficulty in Settling Small Transactions

Peer-to-peer energy trades are small in size and high in number. Traditional banking and billing systems are too costly and slow to settle such micro-transactions efficiently.

2.7. Inefficient Use of Local Generation

Energy generated on one rooftop may be needed only a few houses away, yet it travels through the wider grid with transmission losses. Local sharing between houses can reduce these losses and the load on the grid.

2.8. Limited Participation of Households

Most households are only passive consumers and have no simple platform to offer or request energy. The absence of easy-to-use applications discourages households from investing in renewable generation.

2.9. Weak Auditability for Grid Operators

Grid operators need reliable data on energy flow to plan capacity and detect abnormal activity. Fragmented and manually maintained records make audits slow and error-prone.

2.10. Lack of Automated Settlement

Payments for shared energy are usually handled later through monthly bills. There is no mechanism that automatically releases payment to the producer as soon as the agreed energy is delivered.

Due to these challenges, there is a growing need for a secure, transparent and automated energy sharing system. Blockchain technology offers a promising solution by recording energy offers and transactions on a shared ledger, enabling direct trading between houses, and automating settlement through smart contracts.

3. BLOCKCHAIN BASICS

Blockchain technology is a distributed ledger system that ensures secure, transparent, and tamper-proof transactions. Unlike traditional systems that rely on intermediaries, blockchain enables trustless peer-to-peer interactions, reducing the need for third parties in financial transactions, supply chain management and energy trading. Below are the key concepts that define blockchain technology:

3.1. Decentralization

Traditional databases are managed by centralized authorities, which creates a single point of failure and makes them vulnerable to hacks or data manipulation. Blockchain operates on a decentralized network where multiple independent nodes validate and store data. This ensures that no single entity has full control over the system, making it resistant to censorship and fraud. Since all transactions are recorded and verified across multiple nodes, blockchain also enhances transparency, allowing participants to independently verify transactions. [1]

3.2. Immutability

One of blockchain's defining characteristics is immutability, meaning that once a transaction is recorded, it cannot be altered or deleted. This is achieved through cryptographic hashing, where each block is linked to the previous one, forming a secure chain that prevents unauthorized modifications. Since tampering with a single block would require modifying all subsequent blocks, fraud and unauthorized changes become nearly impossible. This makes blockchain particularly useful for audit trails and energy transaction records. [1]

3.3. Transparency

Blockchain transactions are verifiable by the participants of the network. This transparency is ensured by distributed ledger technology (DLT), where every participant has access to an identical copy of the data, eliminating the risk of hidden alterations. In a permissioned energy network, transparency is limited to authorized households, validators and administrators so that sensitive consumption data stays protected. [3]

3.4. Smart Contracts

Smart contracts are self-executing programs deployed on the blockchain with predefined conditions. Once the specified conditions are met, the contract automatically executes the agreed-upon actions, eliminating the need for manual processing. In an energy sharing system, smart contracts can create energy offers, check energy availability, match buyers and sellers, calculate payment and record each transaction without human intervention. [2] [3]

3.5. Consensus Mechanisms

Blockchain networks rely on consensus mechanisms to validate transactions and maintain system integrity. In Proof of Work (PoW), nodes solve computational puzzles to validate blocks, as in Bitcoin. In Proof of Stake (PoS), validators are selected based on the value they stake. In Proof of Authority (PoA), only approved and trusted validator nodes validate transactions, which avoids the heavy mining process and suits controlled, permissioned networks. For a neighbourhood energy network, PoA is a suitable choice because trusted organizations such as the grid operator or a housing society can act as validators.

3.6. Cryptographic Security

Blockchain security relies on cryptographic techniques to protect user data and transactions. Each transaction is authenticated using public and private keys, ensuring that only authorized households can initiate transactions. Hash functions convert data into fixed-length unique codes, making it practically impossible to reverse-engineer the original information. If a household's private key is compromised, an attacker could act on its behalf, so key protection is an essential part of the system.

3.7. Types of Blockchain

Blockchains are classified by who is allowed to participate. For energy sharing between houses, a private or permissioned (consortium) blockchain is suitable because participating households, energy organizations and validator nodes can be controlled. The different types are compared in Table 3.1.

Blockchain Type	Meaning	Typical Use
Public Blockchain	Anyone can participate.	Bitcoin, Ethereum
Private Blockchain	Controlled by one organization.	Company or institutional systems
Consortium Blockchain	Controlled by multiple trusted organizations.	Banks, energy institutions, universities
Hybrid Blockchain	Combination of public and private features.	Systems needing privacy and transparency

Table 3.1: Types of blockchain

By combining these key features, namely decentralization, immutability, transparency, smart contracts and cryptographic security, blockchain provides a robust and efficient foundation for local energy markets. It ensures secure, automated and verifiable transactions, reducing fraud and increasing trust between households.

4. USE CASE OVERVIEW

The use case for a Blockchain-Based Energy Sharing system aims to allow houses to share or trade excess electricity in a secure and tamper-resistant way. A household that generates more electricity than it consumes, for example through solar panels, offers the excess energy to another household that needs it. Important energy-sharing transactions are recorded on a blockchain, providing a traceable transaction history and making unauthorized modification difficult.

4.1. Objectives

The primary objective of this system is to enable direct energy sharing between houses without relying on a central broker. Households with surplus energy can create offers, and neighbouring households can purchase the energy at an agreed price. This allows producers to earn a fair return and consumers to buy cheaper local energy. [2]

A key challenge in conventional energy billing is the lack of trustworthy records. Blockchain's immutability ensures that once an energy transaction is recorded, it cannot be altered or deleted, which builds trust between participating households and prevents billing disputes.

Another important goal is automated settlement. When a consumer buys energy, the smart contract verifies that the offer is active, checks the available quantity and the payment, and then releases the payment to the producer automatically. This removes manual verification and speeds up settlement.

Transparency is another major objective. Authorized households and administrators can view recorded transactions in real time, which eliminates disputes over how much energy was shared and at what price. The system also supports sustainability by encouraging the use of locally generated renewable energy.

4.2. Scope of the System

The blockchain-based energy sharing system focuses on creating, matching, recording and viewing energy transactions between houses. The system includes the following users and their functions, summarized in Table 4.1:

1. **Energy Producers**: Generate electricity, create energy offers and share excess energy.
2. **Energy Consumers**: Request or buy available energy and view their transactions.
3. **Prosumers**: Households that both produce and consume electricity.
4. **Grid Operator**: Monitors energy flow and grid-related information.
5. **Admin**: Manages users and monitors energy transactions.
6. **Blockchain Network**: Validates transactions and maintains tamper-proof blockchain records.

Actor	Main Functions
Energy Producer	Generate electricity, create energy offers and share excess energy.
Energy Consumer	Request/buy available energy and view transactions.
Prosumer	Both produces and consumes electricity.
Grid Operator	Monitor energy flow and grid-related information.
Admin	Manage users and monitor energy transactions.
Blockchain Network	Validate transactions and maintain blockchain records.

Table 4.1: Actors and their functions

4.3. System Architecture

The architecture of the Blockchain-Based Energy Sharing System is illustrated in **Figure 4.1** and consists of the following components:

4.3.1. User Layer

This layer consists of the primary users who interact with the system. (**Figure 4.1 - User Layer**).

- **Energy Producers / Prosumers**:
  - Create energy offers with quantity and price.
  - Share excess energy with neighbouring houses.
- **Energy Consumers**:
  - Browse available offers and request or buy energy.
  - View their purchase history and transaction status.
- **Grid Operator / Admin**:
  - Manage registered households and validator nodes.
  - Monitor energy flow and recorded transactions.

4.3.2. Web Application Layer

This layer acts as the interface between users and the blockchain. (**Figure 4.1 - Web Application Layer**). [3]

- **User Interface (Frontend)**:
  - Allows producers to create and manage energy offers.
  - Enables consumers to view offers, buy energy and check transactions.
- **Backend / API Layer**:
  - Handles user authentication and registration.
  - Facilitates communication between users and the blockchain.

4.3.3. Blockchain Layer

This layer ensures security, transparency and immutability of energy transactions. (**Figure 4.1 - Blockchain Layer**).

- **Smart Contract Layer**:
  - **EnergyTrading Contract**: Stores energy offers and transactions and manages creating offers, buying energy and viewing records.
  - **Smart Contract Execution**: When a producer creates an offer, the contract records it; when a consumer buys energy, the contract verifies the rules and records the transaction.
- **Blockchain Network**:
  - **Distributed Ledger**: Maintains a decentralized, tamper-proof record of all energy transactions.
  - **Authorized Validation (PoA)**: Only approved validator nodes verify transactions and create blocks.

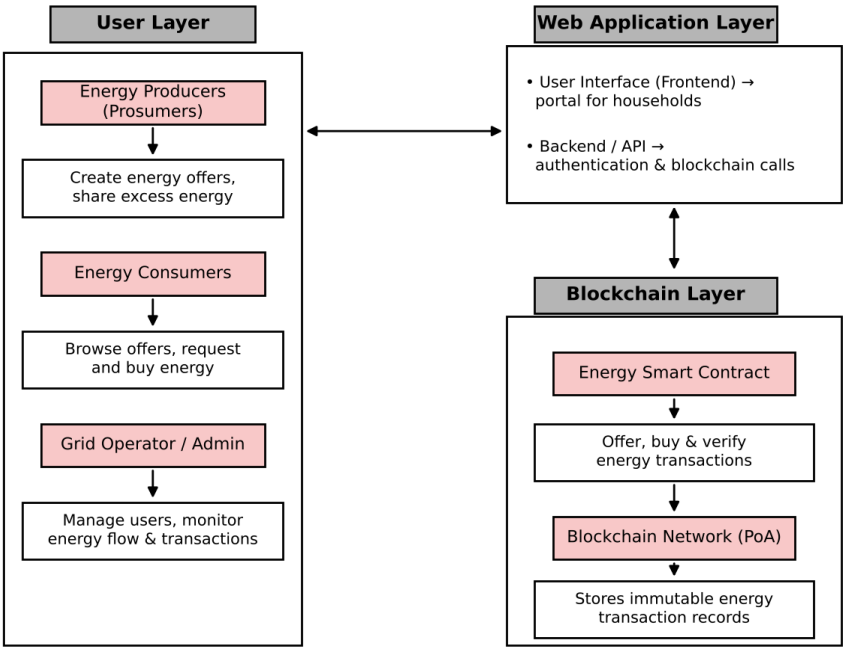


Figure 4.1: Architecture of Blockchain based Energy Sharing System

5. IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

To implement the energy sharing system, the development environment is set up using Ethereum-compatible tools. Truffle serves as the development framework for compiling, testing and deploying smart contracts, and Ganache provides a personal local blockchain for testing. MetaMask is integrated to manage household accounts and sign transactions. The smart contract is written in Solidity (v0.8.19), which provides built-in overflow protection and enhanced error handling. For a permissioned deployment, the same contract can be deployed on a private Ethereum-compatible network that uses Proof of Authority consensus, such as a Geth Clique or Hyperledger Besu network. [3]

5.2. Defining Smart Contracts

5.2.1. Writing the Smart Contract for Energy Trading

The EnergyTrading contract is responsible for registering households, creating energy offers, executing purchases and storing transactions. It defines an EnergyOffer struct holding the offer ID, producer, available energy in kWh, price per kWh, creation time and active status. A second EnergyTransaction struct stores the offer ID, producer, consumer, energy amount, total price and timestamp. The transaction record follows the fields listed in Table 5.1, and mappings link every ID to its stored record.

```
struct EnergyOffer {
        uint256 id;
        address producer;
        uint256 energyKWh;
        uint256 pricePerKWh;
        uint256 createdAt;
        bool active;
}

struct EnergyTransaction {
        uint256 id;
        uint256 offerId;
        address producer;
        address consumer;
        uint256 energyKWh;
        uint256 totalPrice;
        uint256 timestamp;
}

mapping(address => bool) internal registeredHouseholds;
mapping(uint256 => EnergyOffer) internal offers;
mapping(uint256 => EnergyTransaction) internal transactions;
uint256 internal nextOfferId = 1;
uint256 internal nextTxId = 1;
```

```
address public admin;
```

Transaction Field	Description
Transaction ID	Unique identifier of the energy transaction.
Producer ID	Address of the household that supplies the energy.
Consumer ID	Address of the household that buys the energy.
Energy amount	Quantity of energy shared, in kWh.
Energy price	Total price paid for the energy.
Date and Time	Block timestamp of the transaction.
Transaction status	Completed / rejected status of the transaction.

Table 5.1: Fields of an energy transaction

5.2.2. Registering Households

To prevent unauthorized entities from trading, each household must be registered by the admin before it can create offers or buy energy. The function allows registration only once per address and only by the admin, which maintains a verified list of trusted participants. Restricting participation to registered households keeps the network permissioned and prevents fraudulent trading.

```
modifier onlyRegistered() {
        require(registeredHouseholds[msg.sender], "Not registered");
        _;
}

function registerHousehold(address household) external {
        require(msg.sender == admin, "Only admin");
        require(!registeredHouseholds[household], "Already registered");
        registeredHouseholds[household] = true;
}
```

5.2.3. Creating Energy Offers

A registered producer can create an energy offer by specifying the available energy and the price per kWh. The contract checks that both values are greater than zero, assigns a unique offer ID and stores the offer as active. An event is emitted so that the frontend can display the new offer immediately.

```
event OfferCreated(uint256 id, address producer, uint256 energyKWh, uint256 pricePerKWh);

function createEnergyOffer(uint256 energyKWh, uint256 pricePerKWh)
external onlyRegistered returns (uint256) {
        require(energyKWh > 0, "Energy must be > 0");
        require(pricePerKWh > 0, "Price must be > 0");
```

```
        uint256 offerId = nextOfferId++;
        offers[offerId] = EnergyOffer({
                id: offerId,
                producer: msg.sender,
                energyKWh: energyKWh,
                pricePerKWh: pricePerKWh,
                createdAt: block.timestamp,
                active: true
        });
        emit OfferCreated(offerId, msg.sender, energyKWh, pricePerKWh);
        return offerId;
}
```

5.2.4. Buying Energy

A consumer buys energy by calling buyEnergy with the offer ID and the required quantity, and sends the payment with the call. The contract verifies that the offer is active, that the buyer is not the producer, that the requested energy is available and that the payment equals the total price. It then reduces the available energy, records the transaction and transfers the payment to the producer. If any rule fails, the transaction is rejected and the consumer receives an error message. [2]

```
event EnergyTraded(uint256 txId, uint256 offerId, address producer,
        address consumer, uint256 energyKWh, uint256 totalPrice);

function buyEnergy(uint256 offerId, uint256 energyKWh)
external payable onlyRegistered {
        EnergyOffer storage o = offers[offerId];
        require(o.active, "Offer not active");
        require(o.producer != msg.sender, "Cannot buy own energy");
        require(energyKWh > 0 && energyKWh <= o.energyKWh, "Insufficient energy");

        uint256 total = energyKWh * o.pricePerKWh;
        require(msg.value == total, "Incorrect payment");

        o.energyKWh -= energyKWh;
        if (o.energyKWh == 0) { o.active = false; }

        uint256 txId = nextTxId++;
        transactions[txId] = EnergyTransaction(txId, offerId, o.producer,
                msg.sender, energyKWh, total, block.timestamp);
        emit EnergyTraded(txId, offerId, o.producer, msg.sender, energyKWh, total);

        (bool ok, ) = payable(o.producer).call{value: total}("");
        require(ok, "Payment failed");
}
```

5.2.5. Viewing Offers and Transactions

Households and administrators must be able to review offers and recorded transactions. These read-only functions return the stored records by ID and do not cost gas when called from the frontend. This ensures transparency and lets authorized users verify every energy transaction.

```
function getEnergyOffer(uint256 offerId) external view returns (EnergyOffer memory) {
        return offers[offerId];
}

function getTransaction(uint256 txId) external view returns (EnergyTransaction memory) {
        return transactions[txId];
}
```

5.2.6. Cancelling an Energy Offer

A producer may withdraw an offer that is no longer needed, for example when the household's generation drops. The function allows only the producer who created the offer to cancel it and only while it is still active. This prevents other users from tampering with offers and keeps the list of available energy accurate.

```
function cancelOffer(uint256 offerId) external {
        EnergyOffer storage o = offers[offerId];
        require(o.producer == msg.sender, "Not the producer");
        require(o.active, "Offer not active");
        o.active = false;
}
```

5.3. Deploying and Integrating with a Frontend

After developing and testing the smart contract on Ganache, the next step is deploying it to an Ethereum test network (such as Sepolia) or to the private Proof of Authority network. The frontend is developed using React.js, with Web3.js or Ethers.js integrated to interact with the blockchain. Users can register, create energy offers, buy energy and view transactions through a simple and intuitive interface. This ensures that both producers and consumers can share energy without requiring technical blockchain expertise.

5.4. Oracle Integration (Optional Extension)

A blockchain oracle is a mechanism that provides external, off-chain information to a smart contract. In this project, an oracle can supply smart meter readings, energy generated, energy consumed, available excess energy, electricity price and grid status. The flow is: Smart Energy Meter / External Data → Oracle → Smart Contract → Blockchain. If the basic project does not use external meter or grid data, an oracle is not required, so it is treated as an optional

extension. It also addresses the data-accuracy limitation discussed in Chapter 7, because the blockchain cannot by itself know whether a user entered an incorrect quantity.

5.5. Consensus and Block Creation

After an energy transaction is submitted, the blockchain network validates it. Once the transaction satisfies the required rules, it is included in a block, and the block is added to the chain according to the consensus mechanism. The system uses Proof of Authority (PoA), where only authorized validator nodes verify transactions. This avoids the heavy mining of Proof of Work and suits a controlled energy sharing environment in which trusted energy or grid organizations act as validators. The PoA flow is: Energy Transaction → Validator Verifies → PoA Consensus → Block Created → Block Added to Blockchain.

5.6. Workflow:

The workflow of the Blockchain-Based Energy Sharing System, as illustrated in **Figure 5.1**, details the interactions between producers, consumers and administrators to ensure secure offering, buying and recording of energy.

The process begins with the producer household being registered and generating electricity from a renewable source such as solar panels. After meeting its own needs, the household identifies the excess energy and creates an energy offer containing the available energy and price. The offer then becomes available on the platform, where a consumer household can browse it and request the energy. [3]

When a request is made, the smart contract checks the required rules and authorization. As per **Figure 5.1**, if the rules are satisfied, the transaction is validated by the PoA validators, a block is created and the transaction is added to the blockchain, and the payment is released to the producer. If validation fails, the transaction is not recorded as a valid energy transaction and the user receives an error message. Authorized households and administrators can then view the recorded transaction.

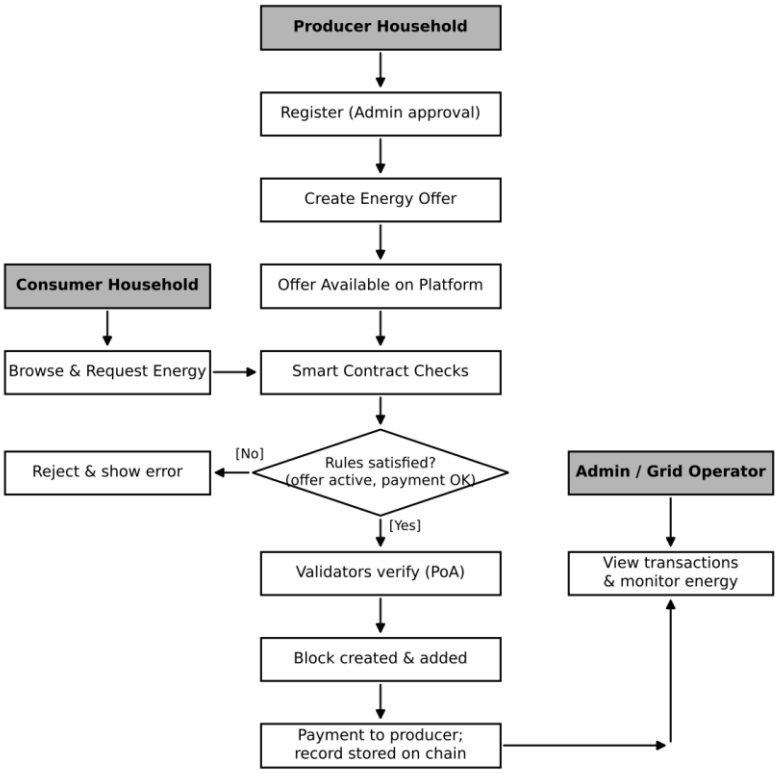


Figure 5.1: Workflow of Blockchain based Energy Sharing System

6. BENEFITS

Blockchain technology enhances energy sharing by introducing transparency, security and automation. Traditional energy trading relies on centralized utilities and billing databases, which are prone to inefficiency, delays and data loss. By leveraging blockchain, energy sharing between houses becomes tamper-resistant, cost-effective and highly efficient. Below are the key advantages:

6.1. Enhanced Security and Immutability

Blockchain provides immutable records where, once an energy transaction is recorded, it cannot be altered or deleted. This prevents fraudulent billing and keeps all transactions tamper-resistant. The cryptographic nature of blockchain secures sensitive data, reducing the risk of hacking and unauthorized modifications. [1]

6.2. Direct Peer-to-Peer Energy Trading

Households can sell surplus energy directly to neighbours without a broker. Producers earn a better price than a feed-in tariff, and consumers pay less than the retail tariff. [2]

6.3. Transparency and Auditability

Authorized participants have real-time visibility of offers and transactions. Since all transactions are recorded on a distributed ledger, any authorized party can verify a transaction without relying on intermediaries, which reduces disputes and increases confidence. [2] [3]

6.4. Automated Settlement through Smart Contracts

Smart contracts verify the offer, quantity and payment, and release the payment to the producer automatically. This removes manual verification, reduces processing time and lowers operational costs.

6.5. Reduced Wastage of Renewable Energy

Surplus energy that would otherwise be wasted is shared with nearby houses that need it. This improves the utilization of rooftop solar generation and reduces transmission losses on the wider grid.

6.6. Cost Reduction and Efficiency

Removing the centralized intermediary and automating verification reduces administrative costs. Small energy payments can be settled quickly without heavy banking or billing overhead.

6.7. Controlled Access with Permissioned Network

Using a permissioned blockchain with Proof of Authority, only trusted households and validator nodes participate. This keeps the network controlled and avoids the high energy use of Proof of Work.

6.8. Encouragement of Renewable Adoption

When households can earn a fair return for sharing energy, investment in solar panels and other renewable sources becomes more attractive, supporting a cleaner and more sustainable energy mix.

6.9. Scalability and Integration with IoT

The system can integrate with smart meters and IoT devices that report generation and consumption automatically through oracles. This makes the process more accurate and reduces manual data entry.

Implementing blockchain in energy sharing brings security, efficiency and transparency to local energy markets. It reduces fraud, lowers administrative costs and encourages the use of renewable energy. The integration of smart contracts and IoT further strengthens the system, making it a future-proof solution for communities that want to manage their own energy.

7. CHALLENGES

While blockchain-based energy sharing offers numerous advantages, it also comes with challenges and limitations that must be considered for effective implementation. Below are some of the key challenges:

7.1. Scalability Issues

A large number of households may generate many small energy transactions, so transaction processing and storage need to be planned. Public blockchains in particular have slow processing speeds and high gas fees, which can become a bottleneck. [3]

7.2. Cost and Resource Requirements

Running blockchain infrastructure, validator nodes, smart meters and related services requires computing resources and initial investment, which may be difficult for small communities.

7.3. Privacy of Household Energy Data

Household energy information is sensitive because it can reveal living patterns. The system must control who can access records, and the design must avoid storing personal details on the blockchain.

7.4. Data Accuracy

Blockchain can make a recorded transaction tamper-resistant, but it cannot automatically know whether an external meter or user supplied incorrect data. Trusted smart meters and oracles are needed to reduce this risk.

7.5. Key and Account Security

If user accounts or private keys are compromised, an attacker may be able to perform unauthorized actions. Households must be educated to store keys securely.

7.6. Regulatory and Legal Compliance

Energy markets are regulated, and rules on selling electricity between households vary across regions. Compliance with energy regulations and data protection laws can make a unified system difficult. [2]

7.7. User Awareness and Adoption

Many households are not familiar with blockchain technology or digital wallets, which leads to resistance to adoption. Simple interfaces and user education are needed.

7.8. Dependence on Internet and Grid Infrastructure

The system relies on internet access and on physical grid or smart-meter infrastructure to actually deliver the traded energy. Areas with poor connectivity may face difficulty using the platform.

Despite these challenges, blockchain-based energy sharing offers transformative potential in enabling local energy markets, enhancing transparency and streamlining settlement. To overcome these limitations, projects must focus on regulatory compliance, user education, security audits and scalable permissioned blockchain solutions.

8. CONCLUSION

This report explores the integration of blockchain technology into energy sharing between houses, addressing key challenges in traditional energy trading. It highlights the inefficiencies of centralized models, such as unfair pricing, wasted surplus energy, manual settlement and lack of transparency. By leveraging blockchain's immutability and decentralization, the proposed system enhances security and trust in energy transactions. The use of smart contracts automates the creation of offers, validation of purchases and settlement of payments, reducing dependency on intermediaries.

The implementation of the blockchain-based energy sharing system, as described in the report, provides a structured approach to trading energy securely. Households can register, create energy offers and track transactions, while consumers can buy energy directly from neighbours. The use of a permissioned network with Proof of Authority consensus keeps the system controlled and efficient. This transition to a blockchain-powered local energy market aligns with the goal of promoting renewable energy, reducing wastage and supporting sustainability.

8.1. Future Outlook for Enhancements

To further optimize the system, smart meters can be connected through oracles to record generated and consumed energy automatically, improving data accuracy. Artificial intelligence (AI) can be used to forecast solar generation and household demand, and to suggest the best time and price to offer energy. AI-driven analytics can also enhance fraud detection by identifying unusual trading patterns. [1]

Integrating decentralized identity (DID) solutions could enhance user security and privacy, ensuring that only verified households can trade. Integration with battery storage and electric vehicles would let households store surplus energy and trade it at peak demand. For wider adoption, interoperability with other energy platforms and utilities can be explored.

Lastly, scalability solutions such as Layer 2 protocols or sharding can be implemented to handle increased transaction volumes efficiently. Dynamic pricing based on real-time supply and demand can make the market fairer. By continuously improving the system with these advancements, blockchain-based energy sharing can evolve into a more secure, efficient and widely adopted solution for community energy.

9. SDG's ADDRESSED

The Blockchain-Based Energy Sharing system aligns with several United Nations Sustainable Development Goals (SDGs) by promoting clean energy, improving infrastructure, supporting sustainable communities and reducing emissions. Below are the key SDGs addressed and their justifications:

SDG 7: Affordable and Clean Energy

The system encourages households to generate and share renewable energy such as solar power. By allowing neighbours to trade surplus energy directly, it makes clean energy more affordable and improves access to it, while reducing wastage of locally generated electricity.

SDG 9: Industry, Innovation, and Infrastructure

Blockchain-based energy sharing promotes digital innovation in the energy sector. Smart contracts, smart meters and tamper-resistant records improve the reliability and efficiency of energy infrastructure and support modernization of local grids.

SDG 11: Sustainable Cities and Communities

Local energy sharing helps neighbourhoods become more self-reliant and resilient. Communities can manage their own energy transparently, which supports sustainable urban living.

SDG 13: Climate Action

By increasing the use of renewable energy and reducing transmission losses and dependence on fossil-fuel based electricity, the system contributes to lower carbon emissions. Automating settlement also reduces paperwork and manual billing processes.

By integrating blockchain, the energy sharing system aligns with these SDGs, fostering a secure, transparent and sustainable ecosystem for managing energy between houses.

10. REFERENCES

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11. APPENDIX A

Project source code and supporting files: https://github.com/Amala0402/VoltChain